

Experiment 1

Title

Hospital Information System for SIMATS Hospital using Zoho Forms (SaaS)

Aim

To create a cloud-based Hospital Information System using the SaaS model for storing and managing patient information online.

Objective

- To understand the SaaS cloud computing model.
- To create an online patient registration form.
- To store patient records securely in the cloud.
- To access patient details from any device with internet connectivity.

Cloud Service Provider

Zoho Forms (SaaS)

Steps to Create the Application

Step 1: Sign Up

- Open **Zoho Forms**.
- Create a free account or sign in.

Step 2: Create a New Form

- Click **Create Form**.
- Select **Blank Form**.

- Name the form:

SIMATS Hospital Information System

Step 3: Add the Following Fields

Field Name	Field Type
Patient Name	Single Line
Gender	Multiple Choice / Dropdown
Age	Number
Date of Birth	Date
Address	Multi Line
Phone Number	Phone
Email	Email
Blood Group	Dropdown
Disease	Multi Line
Doctor Name	Dropdown or Single Line
Department	Dropdown
Appointment Date	Date
Attender Name	Single Line
Attender Phone	Phone
Emergency Contact	Phone
Admission Type	Dropdown (Inpatient/Outpatient)
Insurance Available	Yes/No

Field Name	Field Type
Patient ID	Auto Number or Single Line
Remarks	Multi Line

Step 4: Form Settings

- Make required fields mandatory.
- Enable email notifications (optional).
- Save the form.

Step 5: Publish

Click **Publish** and copy the public link.

Step 6: Test the Form

Enter sample patient details.

Example:

Field	Sample Data
Patient Name	Rahul Kumar
Gender	Male
Age	28
Address	Chennai
Phone	9876543210

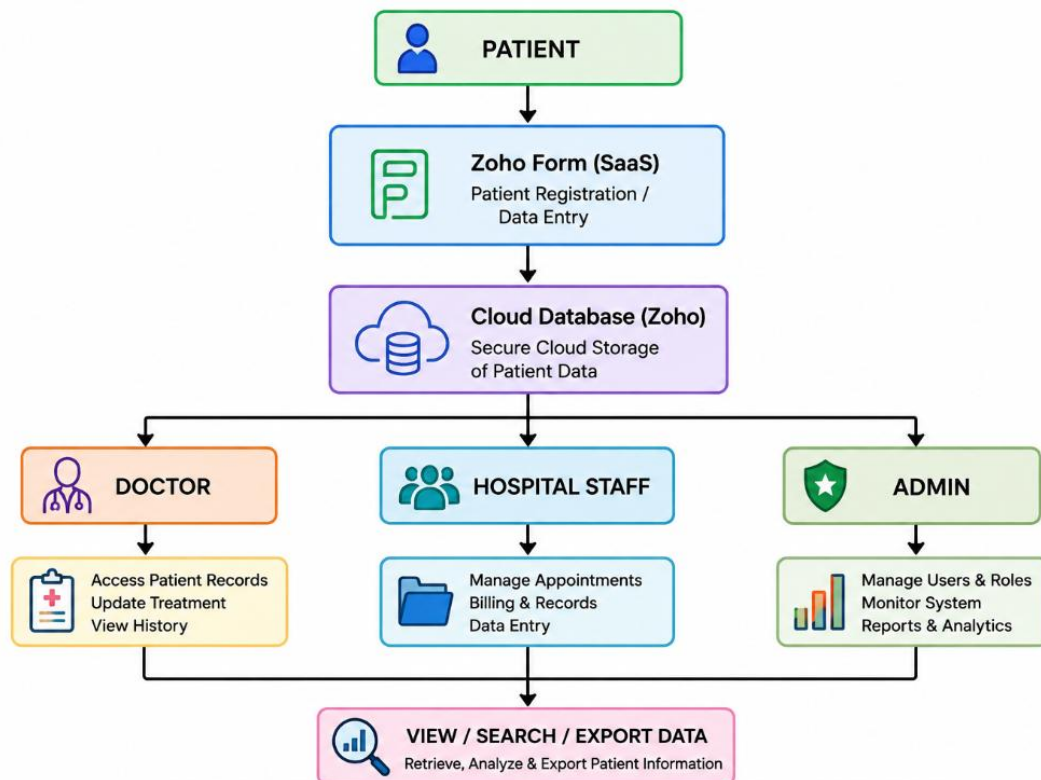
Field	Sample Data
Email	rahul@gmail.com
Attender Name	Ramesh Kumar
Attender Phone	9123456789
Doctor Name	Dr. Priya
Disease	Fever
Department	General Medicine
Blood Group	O+

Submit the form.

Step 7: View Responses

- Open **Responses** in Zoho Forms.
- Verify that the submitted patient data is stored in the cloud.

Architecture Diagram



OUTPUT

SIMATS Hospital Information System Report

☐	Name ▾	Email ▾	Address ▾
☐	Raya	Raya34@gmail.com	Kanchipuram, TN
...	☐ Rahul Kumar	rahul@gmail.com	Kanchipuram, TN

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Edit

Duplicate

More ▾

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Name	Rahul Kumar
Email	rahul@gmail.com
Address	Kanchipuram, TN, India
Phone	+919123456789
Gender	Male
Doctor Name	Dr. Priya
Blood group	O+
DISEASE	Fever
Date Of Birth	04-Aug-2002
APPOINTMENT DATE	04-Aug-2026
ADISSION TYPE	IN PATIENT
INSURANE AVILABLE	YES
REMARKS	NA

Showing 2 of 2

Result

A cloud-based **Hospital Information System for SIMATS Hospital** was successfully created using **Zoho Forms**, demonstrating the **SaaS (Software as a Service)** cloud model. Patient details such as name, age, address, phone number, email, attender details, doctor name, and disease information were collected, stored, and managed online through the cloud.

EXPERIMENT-3

Title

Online Super Market Cloud Software Application using Zoho Forms (SaaS)

Aim

To create a cloud-based Online Super Market application using the SaaS model for collecting customer orders and storing them securely in the cloud.

Objective

- To understand the SaaS cloud service model.
- To create an online supermarket order form.
- To classify items into different categories.

- To store customer orders in the cloud.

Steps to Create the Application

Step 1

Open **Zoho Forms** and log in.

Step 2

Click **Create Form** → **Blank Form**

Form Name:

ONLINE SUPER MARKET

Customer Details

Field Name	Field Type
Customer Name	Single Line
Address	Multi Line
Phone Number	Phone
Email Address	Email

Field	Value
Customer Name	Rahul Kumar
Address	Chennai
Phone	9876543210
Email	rahul@gmail.com
Category	Grocery
Sub Category	Rice
Item Name	Ponni Rice 5kg
Quantity	2
Order Date	04-08-2026
Payment	UPI
Delivery	Home Delivery

Cloud Architecture



OUTPUT



ONLINE SUPER MARKET

Quality You Trust, Service You Deserve

Customer Order Form

Customer Details

Customer Name *

Phone Number *

Email *

Address *

Order Details

Order Date *

Payment Method *

Delivery Type *

Delivery Date *

Item Classification

Category *

Sub Category *

Items and Quantity

Item Name *	Quantity *	Unit Price (₹) *	Total (₹)
Apple - Red	2	150	300
Banana	1	60	60
Orange	1	80	80

+ Add Item

Order Summary

Sub Total (₹)	Delivery Charges (₹)	Discount (₹)	Grand Total (₹)
440	40	20	460

Submit Order

All Responses

View Form

+ Add

📄 Edit

🗑 Delete

📄 Export

🖨 Print

⋮

☐ Show empty fields

☐	Order ID	Customer Name	Phone Number	Items	Total Amount (₹)	Order Date	Delivery Type	Payment Method
☐	ORD-1001	Rahul Kumar	9876543210	Apple, Banana, Orange	460	05/06/2026	Home Delivery	Online Payment
☐	ORD-1002	Anjali Sharma	9123456780	Milk, Bread, Butter	230	05/06/2026	Home Delivery	COD
☐	ORD-1003	Vikram Singh	9988776655	Rice, Dal, Sugar	520	05/06/2026	Store Pickup	Online Payment
☐	ORD-1004	Meena Iyer	9001122334	Aloo, Onion, Tomato	120	06/06/2026	Home Delivery	COD
☐	ORD-1005	Suresh Babu	9887766554	Cooking Oil, Salt	310	06/06/2026	Home Delivery	Online Payment

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ONLINE SUPER MARKET – CLOUD APPLICATION FLOW (SaaS)



CUSTOMER

- Browse Items
- Add to Cart
- Place Order



ONLINE FORM (Zoho Form)

- Customer Details
- Address, Phone, Email
- Item Classification
- Items & Quantity
- Order Summary



CLOUD DATABASE (Zoho)

- Store Orders
- Store Customer Data
- Store Item Details
- Secure Cloud Storage



ADMIN

- Manage Items
- Manage Categories
- View Orders
- Reports & Analytics



STAFF

- Process Orders
- Update Delivery Status
- Manage Inventory



CUSTOMER

- View Order Status
- Order History
- Reorder

Result

The **Online Super Market Cloud Software Application** was successfully created using **Zoho Forms**, demonstrating the **Software as a Service (SaaS)** cloud model. Customer details, item classifications, quantities, and order information were collected through an online form and stored securely in the cloud for easy access and management.

EXPERIMENT -4

Title

Demonstration of Virtualization using a Type-2 Hypervisor

Aim

To demonstrate virtualization by installing a Type-2 Hypervisor and creating a Virtual Machine with **3 CPU cores, 10 GB RAM, and 10 GB virtual hard disk**.

Objective

- To understand virtualization concepts.
- To install a Type-2 Hypervisor.
- To create and configure a Virtual Machine.
- To install a guest operating system.

Requirements

- **Host Operating System:** Windows 11 (or Ubuntu Linux)
- **Type-2 Hypervisor:** VMware Workstation Player / Oracle VirtualBox
- **Guest Operating System:** Ubuntu Linux (or Windows)
- Ubuntu ISO file

Procedure

Step 1

Download and install **VMware Workstation Player** (or **Oracle VirtualBox**).

Step 2

Open the virtualization software.

Step 3

Click **Create New Virtual Machine**.

Step 4

Select the downloaded **Ubuntu ISO**.

Step 5

Enter the VM details.

Setting	Value
Virtual Machine Name	Ubuntu_VM
Guest OS	Ubuntu Linux 64-bit
CPU	3 Processors (3 CPU Cores)
RAM	10 GB (10240 MB)
Hard Disk	10 GB
Network Adapter	NAT
Storage Type	Virtual Disk

Step 6

Click **Finish**.

Step 7

Power on the VM.

Step 8

Install Ubuntu Linux.

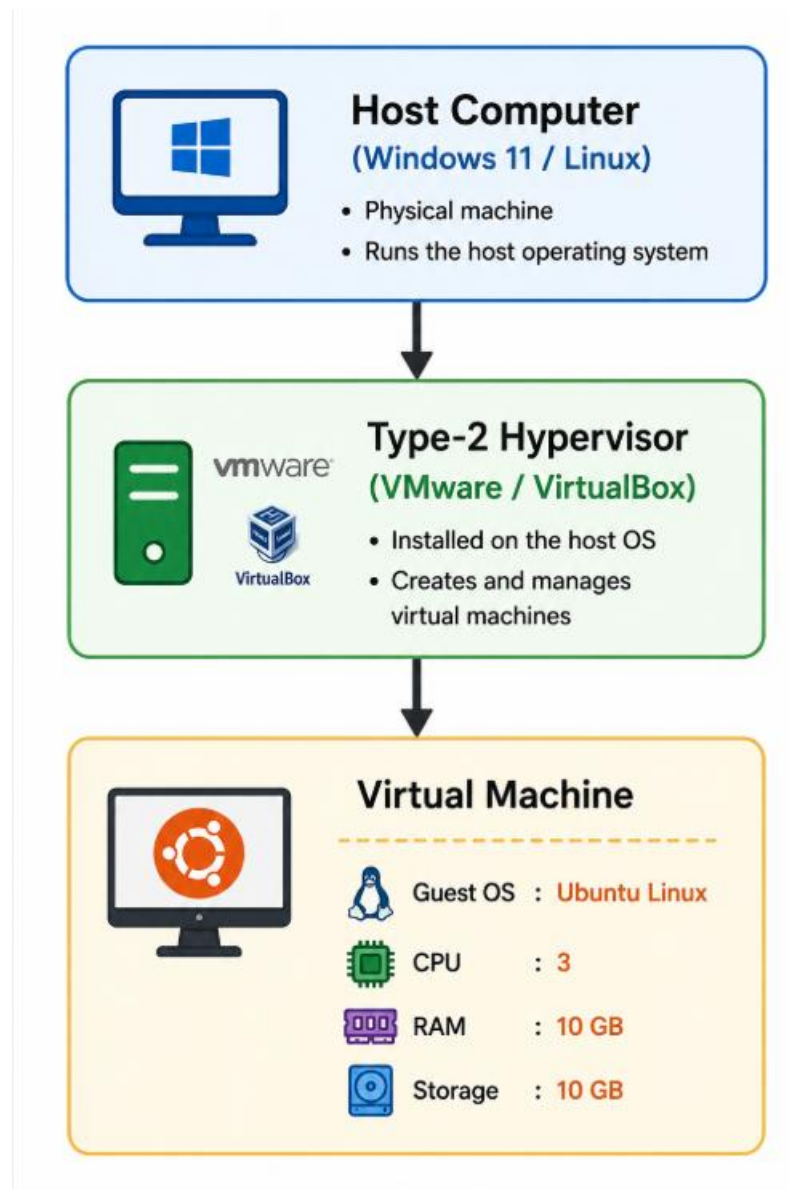
Step 9

Verify the configuration inside the VM.

Virtual Machine Configuration

Parameter	Value
Hypervisor	VMware Workstation Player
Host OS	Windows 11
Guest OS	Ubuntu 22.04 LTS
CPU	3
RAM	10 GB
Storage	10 GB
Network	NAT

Architecture Diagram



OUTPUT

VIRTUALIZATION DEMONSTRATION USING TYPE-2 HYPERVISOR (VMware Workstation Player)

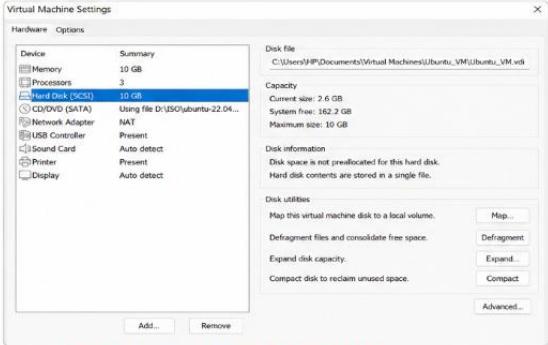
Host OS: Windows 11 Hypervisor: VMware Workstation 17 Player Guest OS: Ubuntu 22.04 LTS
VM Configuration: 3 CPU, 10 GB RAM, 10 GB Storage



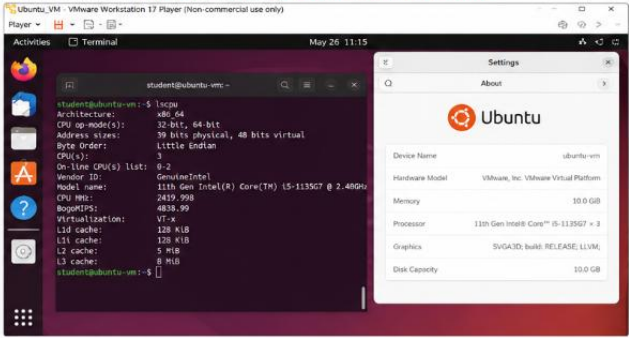
1. VMware Workstation Player Installed on Host OS (Windows 11)



2. VM Creation Summary (3 CPU, 10 GB RAM, 10 GB Disk)



3. VM Configuration – 3 CPU, 10 GB RAM, 10 GB Storage



4. Ubuntu VM Running Successfully with 3 CPU, 10 GB RAM, 10 GB Disk

Result

A **Type-2 Hypervisor** was successfully installed on the host operating system, and a Virtual Machine was created with **3 CPU cores, 10 GB RAM, and 10 GB storage**. The guest operating system (Ubuntu Linux) was installed and executed successfully, demonstrating virtualization.

Experiment 4: Demonstrate Virtualization Using Type-2 Hypervisor

Aim

To demonstrate virtualization by installing a **Type-2 Hypervisor (VMware Workstation/Oracle VirtualBox)**, creating and configuring a Virtual Machine, taking a snapshot, and restoring the VM to a previous state.

Requirements

- Host Operating System: **Windows 11** (or Linux)
- Type-2 Hypervisor:
 - VMware Workstation Player **or**
 - Oracle VirtualBox
- Guest Operating System:
 - Ubuntu Linux ISO

Procedure

Step 1: Install Type-2 Hypervisor

1. Download VMware Workstation Player or Oracle VirtualBox.
2. Install it on the host computer.
3. Restart the system if required.

Step 2: Create a Virtual Machine

1. Open VMware/VirtualBox.
2. Click **Create New Virtual Machine**.
3. Select the Ubuntu ISO file.
4. Configure the VM.

Setting	Value
Guest OS	Ubuntu Linux
CPU	3 Cores
RAM	10 GB
Hard Disk	10 GB
Network	NAT

5. Finish the setup.

Step 3: Install Ubuntu

1. Start the VM.
2. Follow the Ubuntu installation wizard.
3. Complete the installation.
4. Restart the VM.

Step 4: Verify VM

Check:

- Ubuntu boots successfully.
- Internet connection works.
- System settings display:
 - CPU
 - RAM
 - Storage

Step 5: Create Snapshot

1. Shut down (or pause) the VM.
2. Open **Snapshot Manager**.
3. Click **Take Snapshot**.
4. Name it:

Ubuntu_Initial

5. Save the snapshot.

Step 6: Modify VM

Perform any changes such as:

- Create a folder.
- Install software.
- Change wallpaper.
- Delete a file.

These changes will help verify snapshot restoration.

Step 7: Restore Snapshot

1. Open Snapshot Manager.
2. Select **Ubuntu_Initial**.
3. Click **Restore/Revert**.
4. Start the VM.

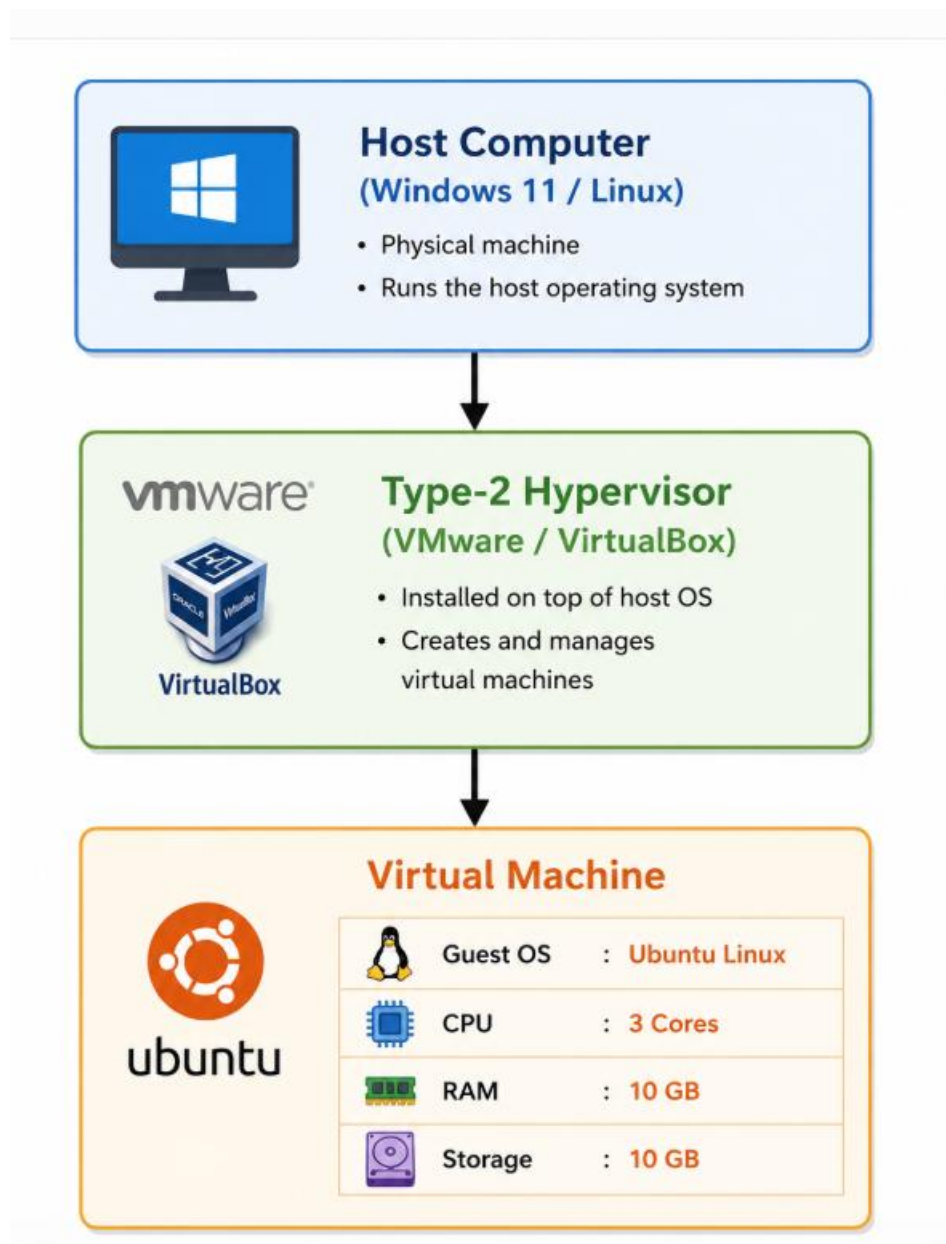
Observe that all changes made after the snapshot are removed.

Step 8: Clone the VM (Optional)

1. Right-click the VM.
2. Select **Clone**.
3. Choose **Full Clone**.
4. Finish cloning.
5. Start the cloned VM.

The clone works independently of the original VM.

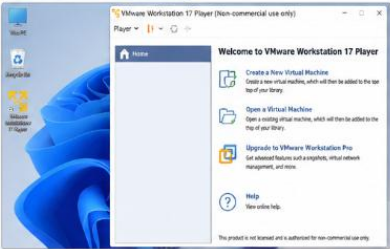
Architecture Diagram

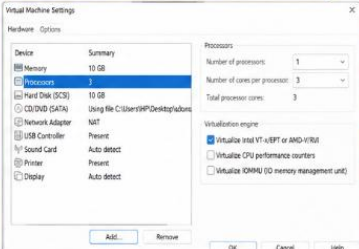


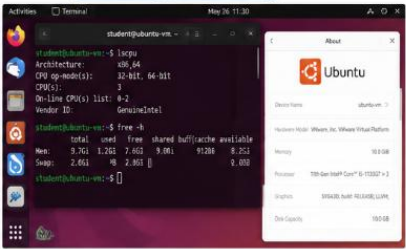
OUTPUT

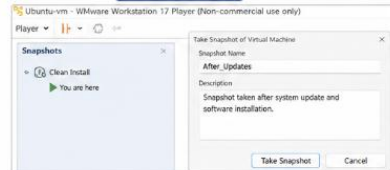
EXPERIMENT 4: VIRTUALIZATION USING TYPE-2 HYPERVISOR – SNAPSHOT AND CLONED VM

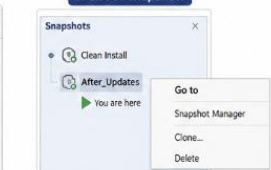
Type-2 Hypervisor: **VMware Workstation Player 17** | Host OS: Windows 11 | Guest OS: **Ubuntu 22.04 LTS**
 VM Configuration: 3 CPU, 10 GB RAM, 10 GB Storage

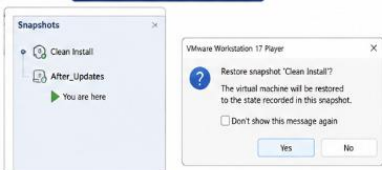
1. Host System (Windows 11)


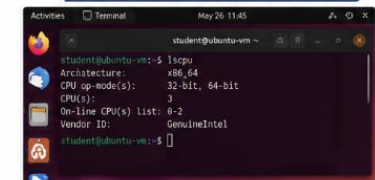
2. VM Configuration (3 CPU, 10 GB RAM, 10 GB Disk)


3. VM Running (Ubuntu 22.04 LTS)


4. Creating Snapshot


5. List of Snapshots


6. Restore to Previous Snapshot


7. VM After Restoring Previous Snapshot (Clean Install)


8. Cloned VM


Observation / Result:

- Snapshot 'After_Updates' created successfully.
- Restored to previous snapshot 'Clean Install'.
- VM returned to the earlier state.
- Cloned VM 'Ubuntu-Clone' created successfully.

Virtualization with Snapshot and Cloned VM tested successfully.

Result

Virtualization was successfully demonstrated by installing a Type-2 Hypervisor, creating and configuring an Ubuntu Virtual Machine, creating a snapshot, restoring the VM to a previous version, and verifying the successful operation of the cloned virtual machine.